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Started	Wednesday, 28 October 2026, 8:04 PM
Completed	Wednesday, 28 October 2026, 8:09 PM
Duration	5 mins 4 secs
Grade	**13.00** out of 15.00 (**86.67**%)

Question **1**

Complete

Mark 1.00 out of 1.00

What is mean by stable sorting algorithm?

a. A sorting algorithm is stable if it preserves the order of all keys

b. A sorting algorithm is stable if it preserves the order of duplicate keys

c. A sorting algorithm is stable if it preserves the order of non-duplicate keys

d. A sorting algorithm is stable if it doesn't preserver the order of duplicate keys

The correct answer is:
A sorting algorithm is stable if it preserves the order of duplicate keys

Question **2**

Complete

Mark 1.00 out of 1.00

How does Merge Sort achieve its efficiency?

a. By using the bubble-up method

b. By comparing elements sequentially

c. By breaking the input into smaller parts and merging them

d. By sorting data in a single pass

The correct answer is: By breaking the input into smaller parts and merging them

Question **3**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm would be preferred for its divide-and-conquer approach?

- a. Linear Search
- b. Merge Sort
- c. Bubble Sort
- d. Binary Search

The correct answer is: Merge Sort

Question **4**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm is known for its simplicity and sequential comparison of elements?

- a. Merge Sort
- b. Heap Sort
- c. Quick Sort
- d. Bubble Sort

The correct answer is: Bubble Sort

Question **5**

Complete

Mark 1.00 out of 1.00

What is one of the first steps in a divide-and-conquer algorithm like Merge Sort?

- a. Dividing the input into smaller subproblems
- b. Combining sorted sublists
- c. Comparing each element with the others
- d. Sorting the entire list sequentially

The correct answer is: Dividing the input into smaller subproblems

Question **6**

Complete

Mark 1.00 out of 1.00

What is the primary advantage of the divide-and-conquer approach in sorting algorithms?

a. It simplifies the sorting process by using only one pass
b. It only works on small datasets
c. It avoids the need for recursion
d. It allows for efficient parallel processing and sorting of data

The correct answer is: It allows for efficient parallel processing and sorting of data

Question **7**

Complete

Mark 1.00 out of 1.00

What type of problems can sorting help solve efficiently?

a. Duplicates
b. Searching
c. Selection
d. All of the above

The correct answer is: All of the above

Question **8**

Complete

Mark 1.00 out of 1.00

Which of the following best describes the term "sorting" in computer science?

a. Arranging data in a specific order
b. Merging two datasets
c. Finding a specific element in a list
d. Removing duplicates from a list

The correct answer is: Arranging data in a specific order

Question **9**

Complete

Mark 1.00 out of 1.00

Which of the following is a key reason for the importance of sorting algorithms?

a. Sorting makes it harder to search for items
b. Sorting helps in finding duplicates quickly
c. Sorting decreases the efficiency of selection operations
d. Sorting is rarely used in programming

The correct answer is: Sorting helps in finding duplicates quickly

Question **10**

Complete

Mark 1.00 out of 1.00

_____ explain how an algorithm will perform when the input grows larger.

a. Searching
b. Sorting
c. Complexity
d. Merging

The correct answer is:
Complexity

Question **11**

Complete

Mark 1.00 out of 1.00

Which built-in Python function is used to sort data?

a. arrange()
b. sorted()
c. order()
d. sort()

The correct answer is: sorted()

Question **12**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm is described as making multiple passes through a list, comparing elements, and swapping adjacent items that are out of order?

a. Insertion Sort
b. Quick Sort
c. Bubble Sort
d. Merge Sort

The correct answer is: Bubble Sort

Question **13**

Complete

Mark 0.00 out of 1.00

Very slow way of sorting is_____

a. Quick sort

b. Insertion sort
c. Bubble sort
d. Heap sort

The correct answer is:
Insertion sort

Question **14**

Complete

Mark 0.00 out of 1.00

Which algorithm is efficient for analyzing the frequency distribution of items in a list?

a. Merge Sort
b. Quick Sort
c. Bubble Sort
d. Linear Search

The correct answer is: Merge Sort

Question **15**

Complete

Mark 1.00 out of 1.00

What is the primary benefit of using sorting algorithms in programming?

a. Decreases the efficiency of algorithms
b. Provides a basis for other algorithms to work efficiently
c. Makes code execution slower
d. Makes data harder to manage

The correct answer is: Provides a basis for other algorithms to work efficiently